

ISOMETRIC

SIDE

TOP

CUTAWAY — internal drive revealed

KEY DIMENSIONS & SPEC

(every number ties back to the flight software)

Shell radius180 mm (Ø360)

ball/params.py quaffle.radius = 0.18 m

Mass0.5 kg

quaffle.mass – light despite its size

Top speed7 m/s

quaffle.max_speed (gentle)

Catch radius0.45 m

quaffle.catch_radius (forgiving)

Lift4 × EDF (internal)

swarm core, soft hover

On catchthrust → 0 in hand

ball/quaffle.py capacitive sense

Scoringhoop optical gate

auto pass-through detection

Safetyno-contact floor

even the nice ball won't bonk you

NOTE

Soft hover, generous catch, rides with its holder, then throws to the nearest hoop with tunable aim-assist (on for youth, off for pros).

BILL OF MATERIALS — QAF (12 line items, prototype qty 1)

REF	PART	SPEC	QTY	\$ EA	\$ EXT
QAF-01	Red grippy foam sphere shell	Ø360 mm (R=180 mm), seamed + finger grooves	1	75	75
QAF-02	Grab-handle ring	recessed leather ring, holds the catch sensor	1	14	14
QAF-03	Ducted-fan lift unit	Ø48 mm internal, internal swarm	4	42	168
QAF-04	Brushless motor	low-speed, smooth	4	20	80
QAF-05	4-in-1 ESC	15 A, telemetry	1	34	34
QAF-06	Flight controller	IMU on-board, throw + aim-assist	1	40	40
QAF-07	Capacitive catch sensor array	detects a holding hand -> thrust to 0	1	18	18
QAF-08	Hoop pass-through beacon	IR tag read by the hoop optical gate	1	10	10
QAF-09	No-contact avoidance compute	even the nice ball won't bonk you	1	0	0
QAF-10	Li-Po pack	3S, long hover endurance	1	22	22
QAF-11	Status LED	warm white possession glow	1	5	5
QAF-12	Telemetry link	reports pose to the Dementor	1	14	14

UNIT TOTAL (prototype):\$480